AI Receptionist R&D

1. Call Answering & Patient Interaction

Example products

• https://www.curemd.com/virtual-medical-receptionist

• https://www.hyro.ai/healthcare/

Features

• 24/7 automated call answering

• intent detection ("book appointment", "refill request", etc.)

• urgent-call routing

• multilingual conversations

• voicemail replacement with AI conversation

AI receptionists can "answer calls, schedule visits, route urgent cases, answer FAQs, process

payments and refills, and document interactions."

1. Core Call Flow (End-to-End Logic)

The workflow can be thought of as a pipeline of stages that turn an incoming call into structured actions for the healthcare system:

1) Call Answering & Greeting

- Incoming patient call is answered instantly (no hold).

- AI greets the caller, often with practice/clinic branding.
- Multilingual support may adapt language automatically.

2) Intent Detection & Context Setting

- The system uses speech recognition + NLP to **identify intent** (e.g., “book appointment,” “refill request,” “urgent issue,” “billing question”).
- Follow-up questions gather necessary details like patient name, date of birth, symptoms, preferred doctor/slot, pharmacy info, etc.

3) Authentication & Lookup

- Caller identity is validated or matched with existing patient records (often via API to EHR/PMS).

4) Task Execution

Depending on intent, the system performs one of several actions:

- **Appointment scheduling** (book/reschedule/cancel)
- **Prescription refills**: collect medication details, verify eligibility, and route to appropriate provider for approval.
- **Insurance verification**: real-time check of coverage/eligibility, co-pays, deductibles.
- **Urgent triage**: detect red-flag terms and escalate to on-call staff or emergency.
- **FAQ answers and general inquiries**: common clinic information without scheduling action.

5) Routing & Escalation

- If the AI detects complexity or an exception, the call is **warm-transferred to human staff** with context summary.
- Urgent/clinical escalation follows predefined protocols.

6) Documentation & Sync

- Structured data, summaries, transcripts, and workflows are logged into the EHR/CRM.
- Follow-ups like SMS/email confirmations or reminders may be sent automatically.

Outcome: a seamless patient experience that integrates call handling, scheduling, triage, and data capture into the clinic's workflows.

2. Appointment Scheduling & Calendar Automation

Example products

• https://www.curemd.com/virtual-medical-receptionist

• https://www.lumahealth.io/

Features

• new appointment booking

• rescheduling & cancellation

• waitlist filling

• no-show recovery calls

• provider availability lookup

Platforms like Luma automate scheduling, referrals, reminders, and waitlists across the patient

Journey.

3. Patient Intake Automation

Example products

• https://www.curemd.com/virtual-medical-receptionist

• https://www.lumahealth.io/

Features

• demographic capture

• consent forms

• visit-reason capture

• pre-visit readiness checklists

• referral intake

AI intake tools sync collected information directly into patient charts before visits.

4. Insurance Verification & Prior Authorization

Example products

• https://www.curemd.com/virtual-medical-receptionist

• https://www.lumahealth.io/

Features

• eligibility verification

• benefits confirmation

• payer intelligence workflows

• prior-auth automation

• copay calculation

Modern systems verify coverage during scheduling and reduce claim rejections automatically.

5. Patient Messaging & EngagementExample products

• https://www.lumahealth.io/

• https://www.hyro.ai/healthcare/

Features

• SMS reminders

• two-way patient chat

• multilingual communication

• follow-up outreach

• automated recall campaigns

Conversational AI messaging improves engagement and reduces manual call workload.

6. Prescription Refill Requests

Example products

• https://www.curemd.com/virtual-medical-receptionist

Features

• refill intake

• medication verification

• provider escalation if required

• pharmacy routing

AI systems can safely process routine refill requests automatically.

7. Payments & Billing Support

Example products

• https://www.curemd.com/virtual-medical-receptionist

• https://www.lumahealth.io/

Features• copay reminders

• payment links

• outstanding balance collection

• eligibility-linked billing workflows

Eligibility checks and payment automation accelerate revenue collection.

8. Referral & Fax Workflow Automation

Example products

• https://www.lumahealth.io/

Features

• referral intake

• inbound fax reading

• specialist scheduling

• care-gap closure workflows

New AI workflow engines even extract findings from faxed clinical reports automatically.

9. Conversational AI Knowledge Assistant

Example products

• https://www.hyro.ai/healthcare/

Features

• clinic FAQs

• provider availability

• accepted insurance plans

• directions & logistics

• service explanations

These assistants resolve large volumes of routine calls automatically.

10. EHR Integration (Critical Feature)

Example products

• https://www.lumahealth.io/

Features

Integration with:

• HealUS

Healthcare AI receptionists sync scheduling, intake, and patient data directly into EHR systems.

11. Analytics & Workflow Reporting

Example products

• https://www.curemd.com/virtual-medical-receptionist

Features

• call metrics dashboards

• booking conversion tracking

• patient interaction analytics

• operational efficiency reports

Dashboards help clinics optimize workflows and staffing decisions.

How an AI Receptionist Handles Payments

1. The AI Listens for Payment Requests

When a patient calls and mentions something about bills, co-pays, outstanding balance, or payments, the AI recognizes that the caller wants to handle a payment — not just book an appointment or ask a question.

This is thanks to its language understanding — it can tell the difference between "I want to book an appointment" and "I need to pay my bill."

https://deepcura.zendesk.com/hc/en-us/articles/46320431931540-AI-Billing-Collect-Payments-with-Your-AI-Receptionist-Scribe-More?utm_source=chatgpt.com

2. The AI Knows the Amount They Owe

The AI doesn't guess the payment amount — it gets that from the clinic's list of services and prices that has been set up ahead of time, or directly from the clinic's payment system (like Stripe).

So it can say something like:

"Your office visit co-pay is $30 — would you like to pay that now?"

This means the patient knows how much they need to pay before any action happens.

3. The AI Sends a Secure Payment Link

Once the patient agrees to pay:

The AI sends a secure payment link by text message (SMS) to the patient’s phone.

That link opens a secure online payment page (handled by payment platforms like Stripe or Square).

On that payment page, the patient enters their card details and completes the payment — just like when you pay for something online or on your phone.

The AI never stores credit card numbers itself — the payment platform does, so it stays secure.

4. The AI Confirms the Payment

Once the patient pays:

The AI gets notified that the payment was successful.

It can send a confirmation message or say that the payment was received.

It can also record that payment in the clinic’s records so the clinic staff doesn’t have to do it manually.

This works the same whether it’s a deposit, co-pay, balance due, or invoice payment.

5. Payment by Invoice (Optional)

If a patient can’t pay immediately via link, the AI can:

Ask for the patient's email,

Generate a formal invoice, and

Send it by email so the patient can pay later with the same secure link.

This option is useful if the patient needs a receipt for insurance reimbursement or employer purposes.

https://usehello.ai/pages/after_hours_revenue_report.html

1. The AI Listens and Decides if It Needs Help

When a patient calls, the AI tries to answer and resolve the question. But not every call can be handled automatically. The AI has *rules* (built into its system) that tell it when it should pass the call on. These rules might include things like:

- The caller says "I want to speak to a human."
- The question is *too complicated* for AI to answer correctly.
- The AI is not confident in its own understanding of what the caller means.
- The call sounds *urgent* or potentially clinical (for example, mentioning serious symptoms).
- The AI has asked three times but still can't resolve the issue.

These are commonly used triggers for escalation.

When one of these *trigger conditions* happens, the AI decides it should escalate. It does **not hang up or dismiss the caller.** Instead, it prepares to transfer the live call.

2. The AI Prepares the Human Transfer

Before the actual hand-off happens, the AI will often:

- **Summarize the conversation so far**, including the reason for the call and any important details the caller already shared.
- **Package that summary** so the human staff doesn't have to ask the same questions again.
- Confirm with the caller that a transfer is happening and who will pick up next ("I'm connecting you with our nurse/staff now").
  This makes the transition smoother for both sides.

3. The AI Does a Warm Handoff (Live Transfer)

Once ready:

1. The AI activates the **transfer action** inside its phone system.
2. The system connects the caller's phone call to a real human staff number or department phone line.

3. The connection is live — the caller stays on the same call without having to hang up and redial.
   This is called a **warm transfer** because the AI *introduces context* and then hands the call over.

You can think of it like this:

- The AI answers the call and listens.
- It catches a cue that says “this needs a real person.”
- It tells the system to dial the staff member’s phone.
- It bridges your call to that person while keeping the same connection.

From the caller’s point of view, the call doesn’t end or restart — it *transitions* to a human agent seamlessly.

4. If No Human Is Immediately Available

The system will fall back on defined logic, such as:

- Leave the caller in a queue (waiting for the next available staff).
- Offer to take a voicemail message instead.
- Ask if the caller wants an SMS/text or callback later.
- Route the call to an after-hours number if it’s outside business hours.
  These are **predefined escalation rules** set by the clinic so nothing gets lost.

Why This Matters in Healthcare

In a medical context:

- Timely escalation can be *critical* if the caller describes urgent symptoms.
- A warm handoff saves time because the staff already knows why the call was escalated.
- Only the calls requiring *clinical judgment or human empathy* get transferred.
  This improves safety, reduces errors from misrouting, and saves staff time

A Simple Example of What Happens

1. **Patient calls.**
2. AI answers and asks, "How can I help you?"
3. Patient says, "I have chest pain."
4. AI recognizes keywords that signal urgency.
5. AI says, "I'm going to connect you to a nurse now."
6. AI transfers the call live to the nurse's phone — with a short summary of what the patient said.

In Summary (Everyday Words)

- The AI listens during the call.

- It has rules that tell it *when it can handle things* and *when it should pass to a human*.
- If the call is complicated or urgent, the AI hands the call to a person.
- Before handing off, it collects key information so the human doesn't have to start from scratch.
- The transfer happens *live on the same call line* so the patient isn't dropped or disconnected.

https://uirix.com/en/voice-in-browser

https://www.plivo.com/docs/voice/use-cases/transfer-to-human-agent

https://www.ionos.com/help/office/ai-receptionist/general-information/configuring-call-transfers/?utm_source=chatgpt.com